

Social Network and Media Analysis

Ch2: Network Models I

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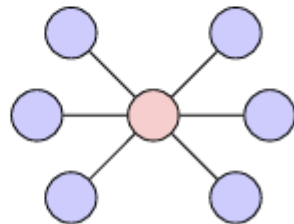
Some slides are excerpted from this lecture: <https://ocw.mit.edu/courses/1-022-introduction-to-network-models-fall-2018/pages/lecture-notes/>

Outline

1. Notion of Centrality
2. Degree
3. Closeness
4. Betweenness
5. Eigenvector
6. Katz

The notion of centrality

- ▶ Identify the **most important nodes** in a graph given its **topology**
 - ⇒ **Not** based on the **nature** of the particular node
- ▶ Different definitions of **importance** give rise to different **centrality measures**
 - ⇒ Degree, closeness, eigenvector, betweenness, Katz
 - ⇒ They induce a **centrality ranking** on the nodes
- ▶ Centrality measures are **widely used**
 - ⇒ Targeted marketing
 - ⇒ Network vulnerability to attacks
 - ⇒ Epidemiology control
 - ⇒ Power in exchange networks



1. Degree centrality

- ▶ **Local** measure of the importance of a node within a graph
- ▶ Sum of the **weights of incident edges**

$$C_D(i) := \sum_{j|(i,j) \in E} A_{ij}.$$

- ▶ **High degree centrality** value of a given node
 - ⇒ The node has a **large number of neighbors**
 - ⇒ **Closely related** to its neighbors (in weighted similarity graphs)
- ▶ For directed networks, both in-degree and out-degree centralities
- ▶ Does not capture **cascade effects**
 - ⇒ I am more important if my neighbors are important

2. Closeness centrality

- ▶ How fast **information can spread** from one node to every other node
- ▶ Inverse of the **sum of the shortest path lengths s_G** to other nodes

$$C_C(i) := \left(\sum_{j \in V} s_G(i, j) \right)^{-1}.$$

- ▶ Nodes which are close to others on average have high centrality
 - ⇒ Such nodes may have more direct influence on others
- ▶ **Example:** Network of movie actors: two actors are connected if they work together
 - ⇒ Highest centrality 0.4143 for Christopher Lee; Entered into the Guinness Book of World Records in 2007 for most screen credits
 - ⇒ Lowest centrality 0.1154 for Leia Zanganeh, an Iranian film and theater actress
- ▶ Two limitations associated with closeness centrality
 - ⇒ Spans a very small dynamic range
 - ⇒ In disconnected graphs, centrality is zero for all nodes

3. Betweenness centrality

- ▶ Control on the **optimal flow** within a graph
 - ⇒ Nodes that fall within the **shortest path** of many pairs of nodes
- ▶ The **number of shortest paths** from j to k is σ_{jk}
- ▶ The **number of shortest paths** from j to k **going through** i is $\sigma_{jk}(i)$

$$C_B(i) := \sum_{\substack{j,k \in V \\ j \neq i \neq k}} \frac{\sigma_{jk}(i)}{\sigma_{jk}}. \quad (1)$$

- ▶ Look at the **shortest paths** for every two nodes distinct from i
 - ⇒ Sum the **proportion that contain node i**
- ▶ Betweenness is not a measure of connectivity.
 - ⇒ For the network of movie actors, highest centrality node is Spanish actor Fernando Rey, with centrality of 7.47×10^8
 - ⇒ He starred in many French, American and Spanish movies.

Interpretation in a social network

- ▶ What would **degree**, **closeness**, and **betweenness** centrality reveal?
- ▶ **Degree** $\Rightarrow$ Most friends $\Rightarrow$ Most **popular** person
- ▶ **Closeness** $\Rightarrow$ Can quickly reach the whole group (directly or indirectly)
 $\Rightarrow$ Relevant if we want to **quickly spread information** in the network
- ▶ **Betweenness** $\Rightarrow$ Power in the transmission of information
 $\Rightarrow$ Relevant if we want to **influence communication** between groups
- ▶ All of them are right, they just reveal different features

4. Eigenvector centrality

- **Linear combination** of the centrality of the neighbors

⇒ **Few important** neighbors can weigh more than **many unimportant** ones

$$C_E(i) := \frac{1}{\lambda} \sum_{(i,j) \in E} A_{ij} C_E(j),$$

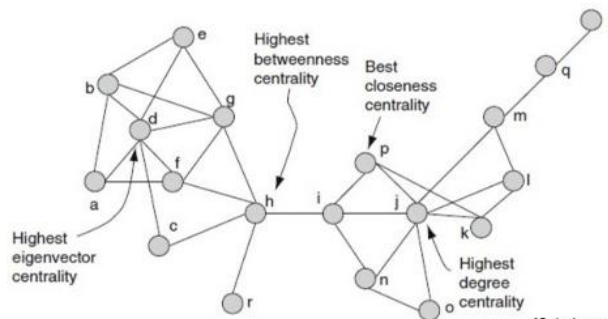
- This definition leads to an **eigenvector** problem
- C_E is the **dominant eigenvector** of the adjacency matrix
⇒ Perron-Frobenius guarantees **all positive elements**

Example

$$A = \begin{pmatrix} 0 & 1 & 1 \\ 1 & 0 & 0 \\ 1 & 0 & 0 \end{pmatrix}$$

Comparing centrality measures

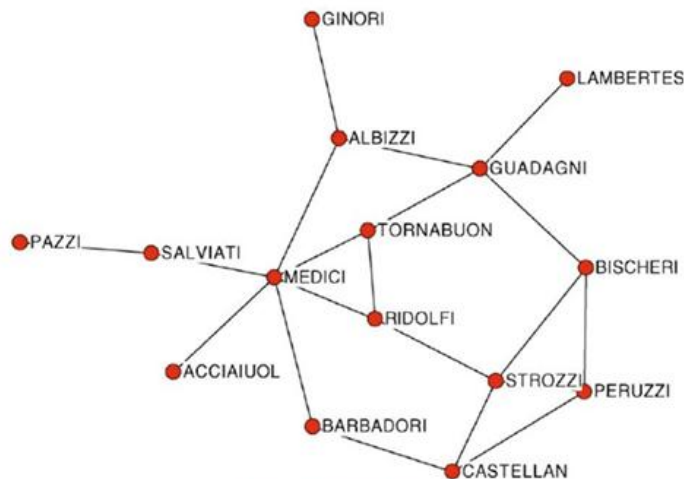
- ▶ Different measures target different notions of importance
- ▶ In a friendship network, degree centrality would correspond to who is the most popular kid.
- ▶ Closeness centrality would correspond to who is closest to the rest of the group,. Useful for spreading information
- ▶ Betweenness would tell us about graph “cut points”, edges whose deletion will cause multiple connected components



[Ortiz-Arroyo 10]
Ortiz-Arroyo, Daniel. "Discovering Sets of Key Players in Social Networks." Chapter 2 in *Computational Social Network Analysis: Trends, Tools and Research Advances*. Springer, 2009. © Springer. All rights reserved. This content is excluded from our Creative Commons license. For more information, see <https://ocw.mit.edu/help/faq-fair-use/>.

Centrality and power in networks

- ▶ Why were the Medicis the most influential family in 15th c. Florence?
- ▶ Political and friendship structure [Padgett & Ansell 93]



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Source: Stephen P. Borgatti. "Centrality and Network Flow," *Social Networks* 27 (2005): 55–71.

- ▶ Highest betweenness centrality by far
⇒ Part of many deals between families supported by marriage linkages

5. Katz centrality

- ▶ Degree centrality only depends on the one-hop neighbors of a node
 - ⇒ One-hop neighbors more relevant than two-hop neighbors
 - ⇒ But indirect relationships are still relevant
- ▶ $[\mathbf{A}^k]_{ij}$ contains the number of walks from i to j of length k
 - ⇒ Consider the degrees of $\mathbf{A}$, $\mathbf{A}^2$, ..., with a discount factor

$$C_K(i) := \sum_{k=1}^{\infty} \alpha^k \sum_{(i,j) \in E} [\mathbf{A}^k]_{ij} = [(\mathbf{I} - \alpha \mathbf{A})^{-1} \mathbf{1}]_i$$

- ⇒ where α is small enough to ensure that the series converges
- ▶ Katz centrality as a hybrid between degree and eigenvector
 - ⇒ Parameter α controls this transition

Use case: a social media and viral marketing campaign

- To maximize the impact of a social media and viral marketing campaign, marketers map the social network as a graph where **nodes are users** and **edges are interactions** (follows, retweets, mentions, or tags).

Goal	Centrality	Target
Mass immediate reach	Degree Centrality	Megaphone (Celebrity)
Connecting two target markets	Betweenness Centrality	Bridge (Cross-over Niche Creator)
Fastest network-wide coverage	Closeness Centrality	Hub (Trend Hopper)
Premium industry credibility	Eigenvector Centrality	Authority (The Expert's Expert)
Long-chain viral loops	Katz Centrality	Sparker (Community Catalyst)

Code

- <https://colab.research.google.com/drive/1nx8ZfVg0KCQU15pCBavmH4g-wbfUkt5Z?usp=sharing>